

OVAR AB, Portugal

WMO: 085440

Lat: 40.9158N

Lon: 8.6411W

Elev: 22

StdP: 101.06

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.0	-2.6	3.0	3.8	-1.0	3.5	3.6	10.1	12.2	9.2	12.6	0.9	0	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	29.2	19.1	27.0	18.6	25.0	18.3	20.9	25.3	20.2	24.1	19.6	23.0	4.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.5	22.4	19.0	13.8	21.4	18.2	13.2	20.7	60.1	26.0	57.8	23.9	55.7	22.9	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.6	7.6	-2.3	34.8	1.3	2.0	-3.3	36.3	-4.0	37.5	-4.8	38.6	-5.7	40.1		
(5)			-2.8	22.3	1.4	1.0	-3.8	23.0	-4.6	23.6	-5.4	24.2	-6.3	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	9.8	10.0	11.8	13.7	15.7	17.7	19.0	19.1	18.3	16.8	12.6	10.3		
	DBStd	4.28	2.64	2.52	2.34	2.56	2.61	2.48	2.09	2.24	2.25	2.75	2.85	2.95		
	HDD10.0	118	38	28	9	2	0	0	0	0	0	0	7	33		
	HDD18.3	1509	265	232	202	142	90	40	13	14	28	62	173	248		
	CDD10.0	1796	31	30	66	113	177	231	279	281	248	212	84	43		
	CDD18.3	144	0	0	0	2	9	20	34	37	27	15	0	0		
	CDH23.3	867	0	0	4	22	81	147	174	198	149	91	1	0		
	CDH26.7	237	0	0	0	4	18	42	68	59	27	17	0	0		
(14) Wind	WSAvg	3.1	2.7	3.1	3.3	3.4	3.5	3.5	3.5	3.3	2.8	2.8	2.6	2.4		
(15) Precipitation	PrecAvg	1260	174	112	100	110	92	38	18	30	78	170	172	166		
	PrecMax	1765	448	318	563	375	219	115	68	151	209	333	414	467		
	PrecMin	682	15	0	0	34	0	0	3	0	9	22	0	0		
	PrecStd	266	112	82	113	80	59	33	15	32	57	76	98	99		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	20.0	23.2	26.0	29.1	30.9	32.2	31.9	29.3	28.8	22.5	19.2		
		MCWB	13.6	13.5	13.9	17.3	18.4	20.0	21.4	20.9	18.4	18.7	16.3	14.3		
	2%	DB	16.0	17.1	20.3	22.8	25.7	27.2	28.1	28.0	27.0	25.8	20.2	17.3		
		MCWB	13.6	12.5	13.5	15.6	17.6	18.4	19.8	18.9	18.3	18.2	15.8	13.8		
	5%	DB	15.1	15.6	18.2	20.2	22.8	24.2	24.1	25.1	24.9	23.6	18.8	16.2		
		MCWB	13.3	12.5	12.8	14.9	16.8	18.0	19.1	18.8	18.1	17.5	15.4	13.6		
	10%	DB	14.2	14.8	16.2	18.2	20.2	21.9	22.3	23.1	23.0	21.8	17.3	15.2		
		MCWB	12.6	12.4	12.7	14.5	15.8	17.5	18.6	18.5	18.3	17.6	15.0	13.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.1	15.6	17.5	19.7	21.0	22.6	22.3	20.9	20.7	18.1	16.1		
		MCDB	16.7	16.3	18.8	23.6	26.3	27.5	30.1	27.4	24.5	24.6	20.0	17.2		
	2%	WB	14.6	14.3	14.7	16.4	18.6	19.6	20.6	20.9	20.1	19.7	17.1	15.1		
		MCDB	15.4	15.7	17.3	20.9	23.6	24.9	25.4	24.6	23.8	22.6	19.1	16.5		
	5%	WB	14.0	13.5	14.1	15.6	17.5	18.7	19.7	20.1	19.4	19.1	16.3	14.4		
		MCDB	14.9	15.2	16.4	19.3	21.4	22.6	23.3	23.2	22.8	21.7	18.1	15.9		
	10%	WB	13.2	12.6	13.4	14.9	16.5	18.0	19.1	19.4	18.7	18.3	15.5	13.6		
		MCDB	14.3	14.5	15.7	18.0	19.5	21.0	21.9	22.2	21.8	21.0	17.0	15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	8.7	8.4	8.3	8.0	7.7	8.8	9.1	9.4	9.1	9.4		
		MCDBR	8.5	10.3	13.2	12.5	13.1	13.3	12.1	12.6	12.9	12.7	10.1	9.2		
	5% WB	MCWBR	6.4	6.9	7.6	6.7	6.4	5.9	5.7	5.6	5.8	6.7	6.5	6.6		
		MCWBR	5.5	7.6	9.0	10.2	10.6	10.4	9.0	9.6	9.7	9.1	7.4	6.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.334	0.345	0.378	0.378	0.385	0.390	0.380	0.393	0.381	0.371	0.343	0.331		
		taud	2.503	2.462	2.375	2.387	2.393	2.397	2.415	2.376	2.431	2.473	2.506	2.509		
	Ebn at Noon	Ebn at Noon	821	866	870	894	892	885	890	868	856	824	803	794		
		Edh at Noon	76	90	110	116	118	117	114	116	103	89	75	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.90	2.93	4.00	5.26	6.42	6.99	7.14	6.45	5.03	3.27	2.15	1.72		
	RadStd	RadStd	0.32	0.41	0.57	0.60	0.57	0.45	0.35	0.31	0.36	0.37	0.36	0.25		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+46

Nomenclature: See separate page